

Incident Response Case Study – SOC Analyst Portfolio (TryHackMe + LetsDefend)

This project is based on simulated lab environments (TryHackMe and LetsDefend), focusing on incident response and threat investigation for portfolio purposes.

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Environment: TryHackMe (Linux/Windows Incident Surface) + Splunk Free (Ubuntu)

Case Type: Lab / Simulated Incident (Training)

1. Executive Summary

A simulated incident investigation was conducted using TryHackMe Linux and Windows hosts, a Windows 10 virtual machine, and a Splunk SIEM deployed on an Ubuntu virtual machine. The analysis identified multiple attacker techniques, including persistence mechanisms (cron jobs, systemd services, and startup scripts), privilege escalation via sudoers modification, credential access configuration (WDigest), and unauthorized remote access tooling (AnyDesk and tunneling).

SIEM analysis in Splunk confirmed visibility into brute-force authentication attempts (Event ID 4625), encoded PowerShell execution, and suspicious file download activity.

No production systems were impacted, as this investigation was performed in a controlled lab environment for training and portfolio development purposes.

Recommended actions focus on removing persistence mechanisms, resetting compromised credentials, improving endpoint logging, and strengthening SIEM alerting.

2. Scope

Systems Analyzed

- TryHackMe Linux host
- TryHackMe Windows host
- Windows 10 virtual machine monitored by Splunk SIEM running on an Ubuntu virtual machine

Data Sources

- Host-based artifacts (users, services, cron jobs, installed packages, filesystem)
- Authentication and system logs (e.g., /var/log/auth.log, dpkg logs)
- SIEM telemetry (Windows Security logs, Sysmon events, PowerShell logs)

- Network traffic capture (PCAP), when available

Time Window

- December 06–07, 2025 (lab activity timeframe)

3. Evidence Collected

3.1 Host Artifacts (Summary)

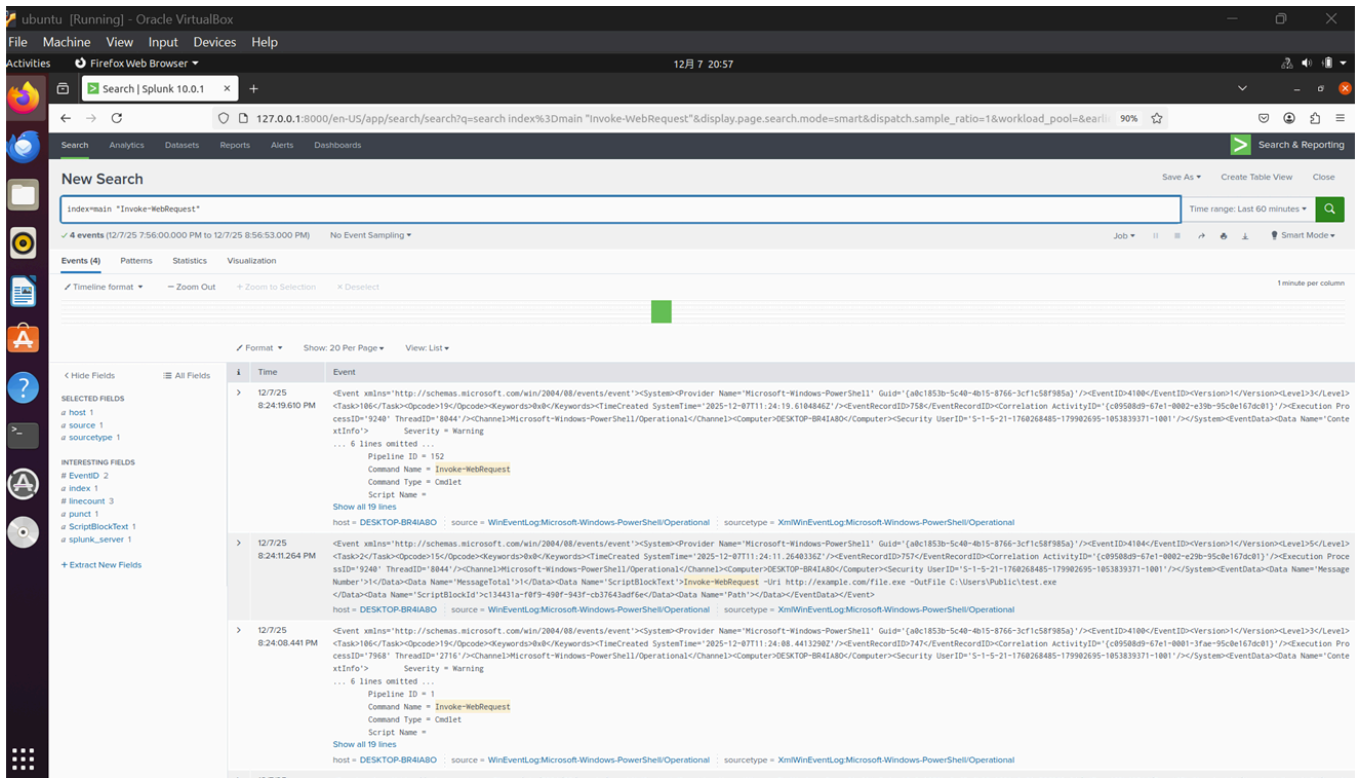
The following artifacts were identified during the investigation across Linux and Windows hosts:

- Suspicious binary executed from a temporary directory: `/tmp/simple`
- Unauthorized local user account created: `attacker`
- Privilege escalation via sudoers modification:
 - `attacker ALL=(ALL:ALL) ALL`
- Persistence mechanisms identified:
 - Cron job executing `/tmp/simple` at system reboot
 - Rogue systemd service: `suspicious.service`
 - Windows startup / scheduled task artifacts
- Suspicious software installation: `malicious-package 1.0`
- Unauthorized remote access tooling:
 - `AnyDesk.exe`
 - Tunneling-related scripts
- Credential access configuration observed:
 - WDigest registry modification (`UseLogonCredential = 1`)
- Disk artifact identified:
 - Hidden NTFS volume

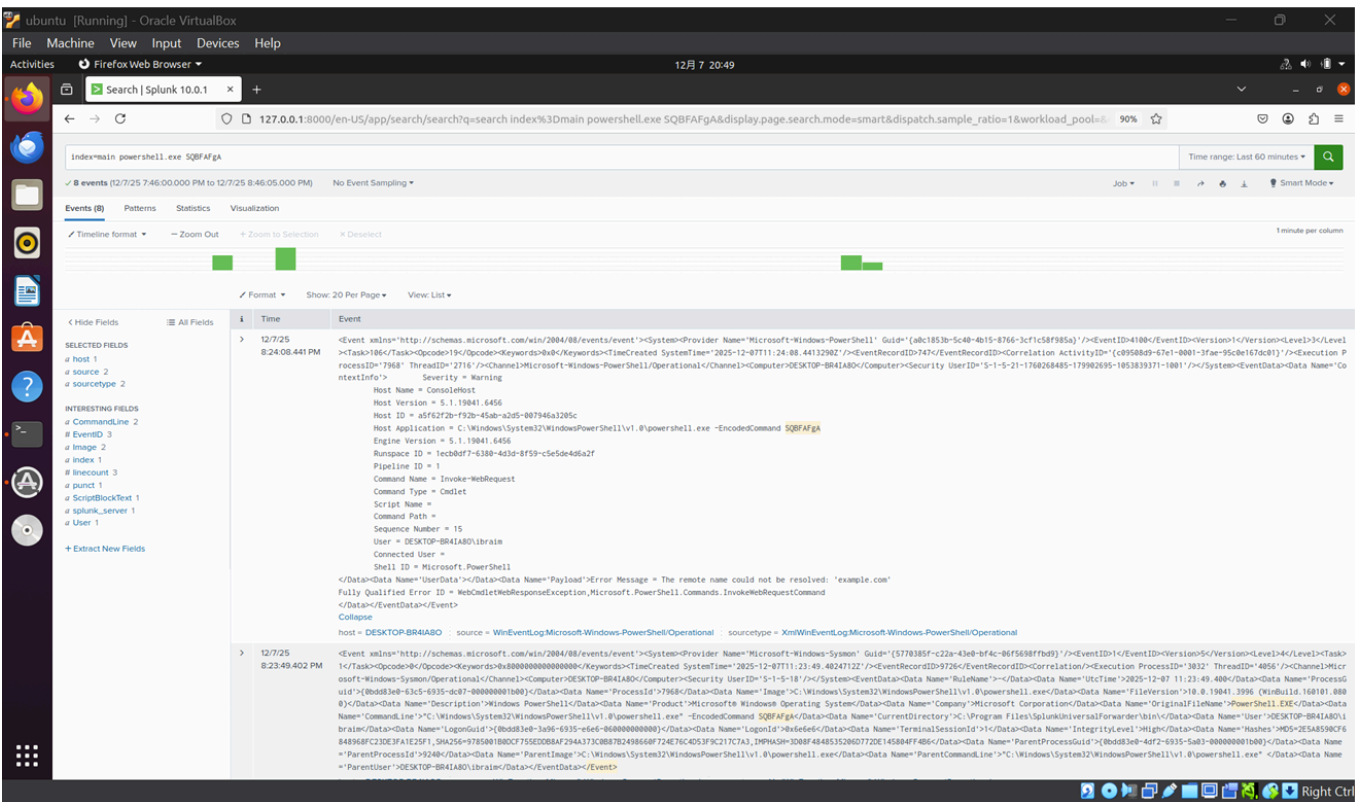
3.2 SIEM Evidence (Splunk)

- Multiple failed logons: EventCode 4625 (brute-force pattern)
- Encoded PowerShell execution: `powershell.exe -EncodedCommand ...`
- Suspicious file download attempt: `Invoke-WebRequest ... file.exe`

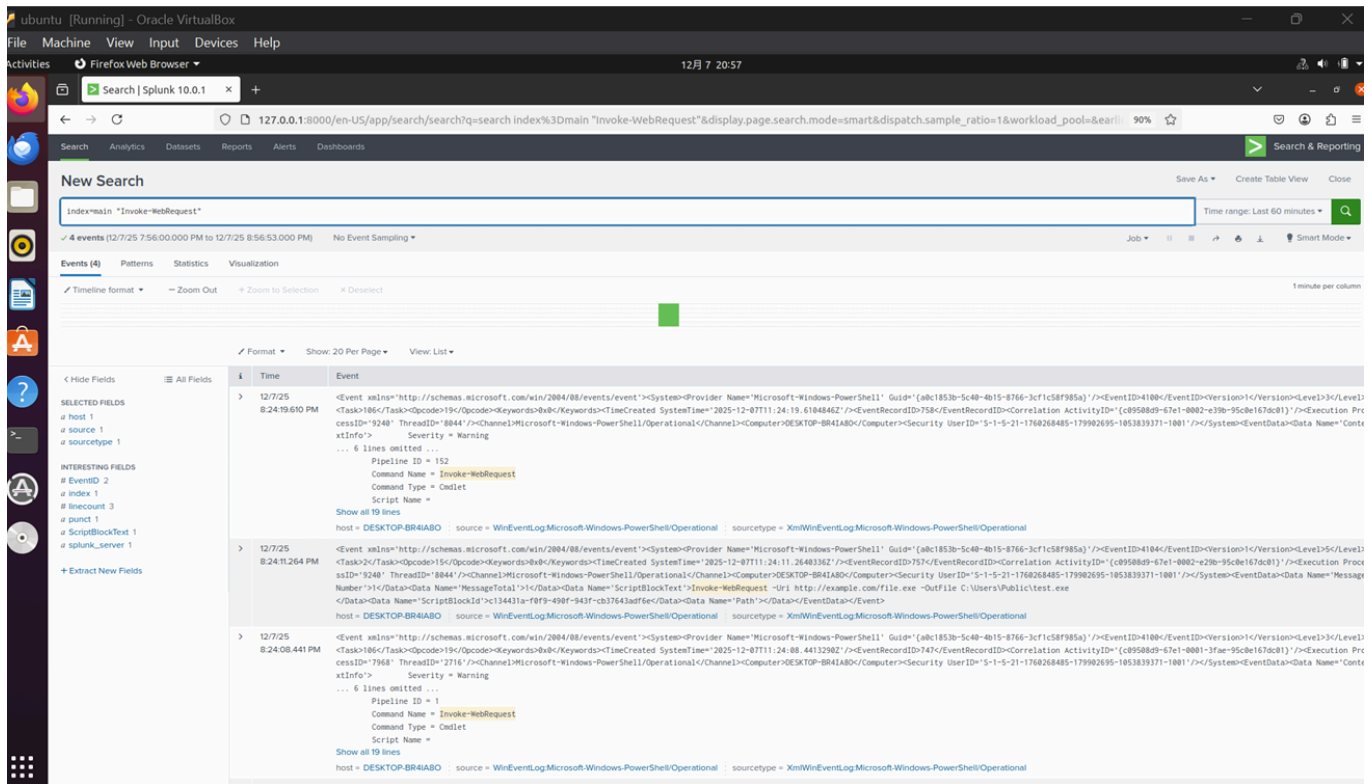
screenshot 1 :Splunk detection showing multiple failed logon attempts (Event ID 4625).



Screenshot 2 — Suspicious PowerShell Encoded Command Execution



Screenshot 3 — Invoke-WebRequest Detection



3.3 Linux Host Evidence (TryHackMe Screenshots)

Screenshot 4 — Malicious Binary Execution (/tmp/simple)

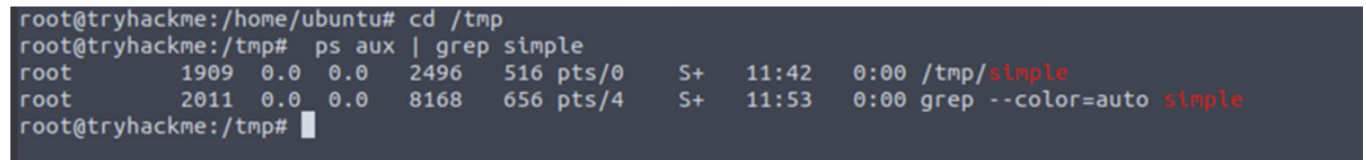


Figure 4: Suspicious binary `/tmp/simple` observed running on the Linux host, confirming active malicious execution.

Screenshot 5 — Unauthorized User and Privilege Escalation

```
File Edit View Search Terminal Help
ubuntu@tryhackme:~$ sudo su
root@tryhackme:/home/ubuntu# sudo useradd attacker -G sudo
root@tryhackme:/home/ubuntu# sudo passwd attacker
New password:
Retype new password:
passwd: password updated successfully
root@tryhackme:/home/ubuntu# echo "attacker ALL=(ALL:ALL) ALL" | sudo
c/sudoers
attacker ALL=(ALL:ALL) ALL
root@tryhackme:/home/ubuntu#
```

Figure 5: Creation of unauthorized user `attacker` with full sudo privileges, indicating privilege escalation.

Screenshot 6 — Cron Persistence Mechanism

```

root ubuntu
root@tryhackme:/var/spool/cron/crontabs# cat root
# DO NOT EDIT THIS FILE - edit the master and reinstall.
# (/tmp/crontab.3EYGAE/crontab installed on Sat Dec 6 12:39:19 2025
# (Cron version -- $Id: crontab.c,v 2.13 1994/01/17 03:20:37 vixie E
# Edit this file to introduce tasks to be run by cron.
#
# Each task to run has to be defined through a single line
# indicating with different fields when the task will be run
# and what command to run for the task
#
# To define the time you can provide concrete values for
# minute (m), hour (h), day of month (dom), month (mon),
# and day of week (dow) or use '*' in these fields (for 'any').
#
# Notice that tasks will be started based on the cron's system
# daemon's notion of time and timezones.
#
# Output of the crontab jobs (including errors) is sent through
# email to the user the crontab file belongs to (unless redirected).
#
# For example, you can run a backup of all your user accounts
# at 5 a.m every week with:
# 0 5 * * 1 tar -zcf /var/backups/home.tgz /home/
#
# For more information see the manual pages of crontab(5) and cron(8)
#
# m h dom mon dow command
@reboot /tmp/simple
root@tryhackme:/var/spool/cron/crontabs#

```

Figure 6: Cron job configured to execute `/tmp/simple` at system reboot, providing persistence.

Screenshot 7 — systemd Persistence Service

```

root@tryhackme:/var/spool/cron/crontabs# sudo systemctl status suspicious.service
● suspicious.service - Suspicious_Service
   Loaded: loaded (/etc/systemd/system/suspicious.service; enabled; vendor preset: enabled)
   Active: active (running) since Sat 2025-12-06 12:47:52 UTC; 30s ago
     Main PID: 2301 (suspicious)
       Tasks: 1 (limit: 4671)
      Memory: 156.0K
        CGroup: /system.slice/suspicious.service
                └─2301 /home/activities/processes/suspicious

Dec 06 12:47:52 tryhackme systemd[1]: Started Suspicious_Service.

```

Figure 7: Malicious systemd service `suspicious.service` enabled and running, ensuring execution at startup.

4. Timeline

4.1 Windows Host Timeline (Splunk)

Time (hh:mm:ss)	Event	Details
20:23:42	Failed Logon Attempt (Event ID 4625)	The account Administrator failed to authenticate on host DESKTOP-BR4IA80 . Reason: Unknown user name or bad password.
20:23:49	Suspicious PowerShell Execution (EncodedCommand)	PowerShell executed using the <code>- EncodedCommand</code> flag, indicating obfuscation and potential malicious script execution.
20:23:49	Suspicious PowerShell Execution (EncodedCommand – second log)	Additional logging confirms the same encoded PowerShell payload executed under user account ibraim , suggesting repeated execution.
20:24:01	Suspicious File Download (Invoke-WebRequest)	PowerShell attempted to download external content using <code>Invoke-WebRequest</code> to <code>http://example.com/file.exe</code> , saving to <code>C:\Users\Public\test.exe</code> , indicating potential ingress tool transfer.

4.2 Linux Host Timeline (Artifact-based)

Order	Event	Evidence
1	Malware executed from /tmp	/tmp/simple running (PID 1909)
2	Privileged rogue user created	user attacker + sudoers modification
3	Cron persistence added	@reboot /tmp/simple
4	systemd persistence added	suspicious.service enabled/running
5	Malicious package installed	malicious-package 1.0 installation evidence

5. Findings + IOCs

5.1 Findings

Linux Host Findings

The Linux host shows clear evidence of post-compromise activity. A suspicious binary (`/tmp/simple`) was executed and maintained through multiple persistence mechanisms, including a cron job and a malicious systemd service. An unauthorized privileged user (`attacker`) was created and granted full sudo access, indicating successful privilege escalation. Additionally, a suspicious package (`malicious-package 1.0`) was installed, suggesting attacker-controlled tooling deployment.

Windows Host Findings

The Windows host exhibited multiple suspicious behaviors detected via SIEM telemetry. These included repeated failed authentication attempts (Event ID 4625), execution of obfuscated PowerShell commands using the `-EncodedCommand` flag, and an attempted external file download via `Invoke-WebRequest`. These activities indicate credential access attempts, script-based execution, and potential ingress tool transfer.

5.2 Indicators of Compromise (IOCs)

IOC	Type	Platform	Description
/tmp/simple	File / Binary	Linux	Suspicious executable running from /tmp directory
PID 1909	Process ID	Linux	Running process associated with malicious binary
attacker	User Account	Linux	Unauthorized privileged user created by attacker
attacker ALL=(ALL:ALL) ALL	Privilege Escalation	Linux	Full sudo privileges granted to rogue user
suspicious.service	systemd Service	Linux	Malicious service configured for persistence
@reboot /tmp/simple	Cron Job	Linux	Persistence mechanism executed on system reboot
malicious-package 1.0	Installed Package	Linux	Suspicious package installed by attacker
AnyDesk.exe	Remote Access Tool	Windows	Unauthorized remote access software
RunWallpaperSetup.cmd	Startup Script	Windows	Persistence via startup execution
WDigest UseLogonCredential=1	Registry Modification	Windows	Enables plaintext credential storage
Invoke-WebRequest	PowerShell Command	Windows	Used to download external executable
http://example.com/file.exe	URL	Windows	External resource used for file download
C:\Users\Public\test.exe	File Path	Windows	Download destination for suspicious executable

6. MITRE ATT&CK Mapping

Activity	Tactic	Technique ID	Technique Name	Evidence
Multiple failed logon attempts	Credential Access	T1110	Brute Force	Windows Event ID 4625 detected in Splunk
Encoded PowerShell execution	Execution	T1059.001	PowerShell	powershell.exe executed with - EncodedCommand
Suspicious file download	Command and Control	T1105	Ingress Tool Transfer	Invoke-WebRequest downloading external executable
Cron-based persistence	Persistence	T1053.003	Scheduled Task/Job: Cron	@reboot /tmp/simple cron entry
systemd service persistence	Persistence	T1543.002	Create or Modify System Process: systemd Service	suspicious.service enabled and running
Sudoers modification	Privilege Escalation	T1548.003	Abuse Elevation Control Mechanism: Sudo	attacker ALL=(ALL:ALL) ALL
Unauthorized remote access tool	Lateral Movement / Persistence	T1219	Remote Access Software	AnyDesk.exe artifact identified
Credential access configuration	Credential Access	T1003	OS Credential Dumping (supporting)	WDigest UseLogonCredential enabled

7. Recommended Remediation

High Priority

- Remove all identified persistence mechanisms (cron jobs, systemd services, startup scripts).
- Delete malicious binaries and scripts, including /tmp/simple and related files.
- Disable and remove unauthorized user accounts and reset compromised credentials.
- Reimage affected systems if full integrity cannot be guaranteed.

Medium Priority

- Add identified IOCs to SIEM detection rules for continuous monitoring.
- Configure alerts for:
 - Multiple failed logon attempts (Event ID 4625)
 - Encoded PowerShell execution
 - Suspicious use of Invoke-WebRequest
- Enable advanced PowerShell logging (Script Block and Module logging).
- Implement account lockout and MFA policies to mitigate brute-force attacks.

8. Final Conclusion

This case demonstrates a realistic attack chain in a controlled lab environment: suspicious execution from temporary paths, privilege escalation, and persistence on Linux (cron and systemd), alongside Windows-based suspicious activity observed via SIEM telemetry. Splunk detections provided visibility into brute-force authentication attempts (Event ID 4625), obfuscated PowerShell execution using EncodedCommand, and attempted external file transfer via Invoke-WebRequest. The collected host artifacts and SIEM evidence consistently support post-compromise behavior and persistence techniques. No production systems were impacted, as this investigation was performed for training and portfolio purposes. Recommended actions focus on eradicating persistence, removing malicious tooling, resetting credentials, and strengthening logging and alerting to improve time-to-detect in similar incidents.

PCAP Findings

PCAP analysis was not provided for this lab; conclusions are based on host artifacts and SIEM telemetry.

9. Skills Demonstrated

- Alert triage and incident investigation
- Log analysis (Windows Event Logs, Sysmon, Splunk)
- IOC identification and threat correlation
- MITRE ATT&CK mapping
- Incident response documentation